MILITARY SPECIFICATION

BUFFING MACHINE
(FLOOR MOUNTED, ELECTRIC)

1. SCOPE

1.1 This specification covers a floor-mounted, single spindle, buffing machine equipped with separate dust hoods for cleaning 5-gallon gasoline cans in drum- and can-reconditioning plants.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

CC-M-641 - Motors, Alternating-Current (Integral-Horsepower, 200 H.P. and Smaller).

Military

MIL-V-173 - Varnish, Moisture-and Fungus-Resistant (for the Treatment of Communications, Electronic, and Associated Electrical Equipment).
MIL-C-1283 - Can, Gasoline, Military 5-Gallon.
MIL-M-18058 - Machinery Metal and Wood-Working, Support Equipment and Associated Repair Parts, Preparation for Delivery of.

STANDARDS

Military

MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of U.S. Military Property

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MIL-STD-461 - Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions, should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated the issue in effect on date of invitation for bids or request for proposal shall apply.

National Fire Protection Association (NFPA)

Standard No. 70 - National Electrical Code.

(Application for copies should be addressed to the National Fire Protection Association, 60 Batterymarch St., Boston, Mass. 02110.)

American Trucking Associations, Inc.

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., 1616 P Street, N.W., Washington, D. C. 20036.)

Uniform Classification Committee

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, Chicago, Illinois 60606.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Description. The buffing machine shall be designed and equipped for the exterior cleaning of 5-gallon cans conforming to MIL-C-1283. The machine shall consist essentially of a pedestal; an electric-motor-driven, double-ended spindle assembly; a push-button stop-start station;

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and two separate, floor-mounted dust collecting hoods. The spindle assembly shall be supported by ball or roller bearings, shall rotate in a downward direction when facing the machine, and shall be driven by a motor mounted integrally with the spindle or by a separately-mounted, V-belt-connected motor. The spindle shall be designed to accommodate the wire brushes specified herein and shall be fully equipped with wheel flanges and safety nuts. The machine shall be provided with means for locking the spindle to facilitate brush removal.

3.2 First article. Unless otherwise specified (see 6.2), the supplier shall furnish, within the time limit specified in the contract or order, a complete buffing machine for first article inspection. The first article may be either a preproduction model or an initial production item which conforms to all requirements of this specification. Inspection of the first article shall consist of the examination and tests specified in 4.3 and 4.4 which shall be conducted prior to the submission for acceptance of the remainder of the buffing machines furnished under the contract. The unit subjected to first article inspection shall be identical to the units the supplier proposes to furnish in fulfillment of the contract. Following acceptance by the Government, the first article may be delivered after required servicing and adjustment as the last unit of supply under the contract. Approval of the first article on the basis of the specified inspection shall not relieve the supplier of his obligation to meet the requirements of the specification for the remaining units delivered under the contract or order.

3.3 Safety standard. Electrical wiring and associated electrical components shall conform to and shall be installed in accordance with the requirements of National Fire Protection Association (NFPA) Standard No. 70.

3.3.1 Certification. Acceptable evidence of meeting the requirements of NFPA No. 70 shall be the contractor's certified statement the buffing machine conforms to the applicable requirements. Compliance with the NFPA standard in regard to safety does not absolve the contractor from complete compliance with the requirements of this specification in order to secure acceptance of this machine.

3.4 Standard product. Except as specified herein, the machine and all its components shall be a commercially developed product of the manufacturer or his suppliers. The items furnished shall be new.

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3.5 Interchangeability. All units shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, and spare parts. No deviations shall be acceptable without prior written approval of the contracting officer.

3.6 Material. Material shall be in accordance with the requirements specified herein. Material not definitely specified shall be of the type and quality normally furnished to produce a unit that meets the requirements described herein, and shall be free from all defects and imperfections that may affect the serviceability and appearance of the finished product.

3.7 Design and performance. The machine shall be provided with an electric motor having a minimum 3 horsepower (h.p.) capacity. Unless otherwise specified (see 6.2), the full load spindle speed shall be approximately 1725 r.p.m. with 12-inch diameter wire brushes. The dimensions of the machine shall include a spindle height of at least 38 inches from the floor, a distance of 10 inches or more from the inside spindle flange to base side, and a distance between brushes of no more than 40 inches. The net weight of the buffer shall not exceed 600 pounds and the machine shall require an approximate floor space not exceeding 6 square feet. All parts subject to wear, breakage, or distortion shall be accessible for adjustment or repair. Design of the machine shall be such as to prevent conditions hazardous to personnel or deleterious to equipment. All externally mounted controls shall be within easy reach of the operator's normal working position. The machine shall be of the manufacturer's latest design, and shall have the capacity to effect exterior cleaning of the cans by removal of paint, rust, scale, varnish and stains.

3.8 Construction. The machine shall be constructed to withstand without permanent damage or deformation, strains, jarring, and vibrations incidental to shipping, storage, installation, and operational service. When installed and connected to the specified power source, its serviceability shall be assured in any operation for which it was designed.

3.8.1 Pedestal. The pedestal shall be a one-piece cored casting of iron or fabricated of heavy gage steel and shall be capable of supporting the entire machine. As applicable, the pedestal top plate shall be designed for attachment of a motor-on-spindle assembly, or both pedestal ends shall accommodate encased spindle extensions and the interior shall receive the mounting of a separate motor and multiple V-belt drive. Machining of the pedestal shall provide for an accurate leveling of the spindle along its longitudinal axis.

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at a height specified herein. Facilities shall permit the lag-screwing or bolting of pedestal to the floor, and shall provide for attachment of removable or hinged access cover(s), preferably of the louvered type.

3.9 Spindle. The spindle shall be a high grade steel, and be precision ground and dynamically balanced for vibrationless operation. Both ends shall be threaded so that spindle nuts shall tighten in the direction of spindle rotation, as described in 3.1, and wheel flanges shall be positioned to meet the capacities specified in 3.7, as applicable. When attachment of a spindle sheave is required, the spindle shall be equipped with a woodruff key.

3.10 Exhaust hood. The exhaust hood shall be fabricated of galvanized steel with an adjustable cover, an adjustable scoop and provided with a 4-inch diameter suction outlet with removable cap and a cleanout door at the bottom front of its base. The exhaust hood shall be capable of accommodating a brush with capacities specified in 3.12.1, at a spindle height as specified herein.

3.11 Electrical wiring and components. Unless otherwise indicated herein, the electrical components shall be designed to operate on 208 volt, 60 hertz, 3 phase alternating current. All connections shall be with the use of terminal boxes.

3.11.1 Switch and magnetic starter. A rigidly mounted explosion-proof magnetic starter, with overload and low voltage protection, shall be provided with a 1/2-inch hub for rigid conduit to the motor. The switch shall be a start-stop push-button type that shall be suitably enclosed for use in hazardous locations.

3.11.2 Motor. The drive motor shall conform to CC-M-641, shall be totally enclosed, equipped with ball bearings, and fan cooled or non-ventilated, as applicable. The horsepower of the single speed motor shall be as specified in 3.7. The motor shall be 3 phase, 60 hertz, alternating current, wired for operation on the voltage specified in 3.11. When required, the motor housing shall be equipped with encased spindle extensions, capable of accommodating ball bearings.

3.12 Accessory equipment.

3.12.1 Standard equipment. The following equipment, as applicable, shall be furnished with each machine:

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a. One spindle nut wrench normally furnished by the manufacturer.

b. Two exhaust hoods, as specified in 3.10.

c. Two steel wire brushes, complete with 1 1/4-inch arbor hole adapters, No. 33 gage (0.0118) wire size, 2-inch operating face width, and trim length of 1 7/8 inches, and of a diameter specified in 3.7.

3.13 Lubrication. All bearings (except sealed-for-life type) and other moving parts, as required, shall be lubricated in accordance with standard commercial practice. Points of lubrication shall be provided with accessible means such as oil holes, grease fittings and filler caps. The unit shall operate satisfactorily when lubricated with standard Military lubricants.

3.14 Treatment and painting. The equipment shall be treated and painted in accordance with the manufacturer's standard practice.

3.15 Identification marking. The contracting officer will furnish the required identification plates. The contractor will be required to stamp necessary data conforming to MIL-STD-130 in the blank spaces of the identification plates, and securely affix said plates in a conspicuous place on the equipment. Brass screws or bolts shall be used to affix the plates. The nomenclature shall be "Buffing Machine."

3.16 Electromagnetic interference controls. When specified (see 6.2), the equipment shall conform to the electromagnetic characteristic requirements and test limits for class No. IIB equipment as specified in MIL-STD-461.

3.17 Fungus control. When specified (see 6.2), electrical connections, including terminal and circuit connections, components, and circuit elements, shall be coated with varnish conforming to MIL-V-173 except that:

a. Components and elements inherently inert to fungi or hermetically sealed need not be treated.

b. Components and elements whose operation will be adversely affected by the application of varnish shall not be treated.

c. One percent of copper 8-quinolinolate (by weight based on the nonvolatile content of the varnish) may, at the supplier's option, be substituted for the salicylanilide specified in MIL-V-173, provided that the varnish conforms to MIL-V-173 in all other respects.

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3.18 Technical publications. Unless otherwise specified (see 6.2), two copies each of the manufacturer's standard operator manual and parts list shall be furnished with each machine. Detail requirements for the technical publications shall be as specified in the contract (see 6.2).

3.19 Spare parts. When required, spare parts shall be as specified (see 6.2), and shall be provided by the manufacturer.

3.20 Workmanship. Workmanship of the machines and accessories shall be of the quality prevailing among manufacturers normally producing equipment of the type specified herein.

3.20.1 Castings and forgings. Castings and forgings shall be free of defects, scale, and mismatching. Repair processes, such as welding, peening, plugging, or filling with cold solder or metallic paste shall not be attempted without authorization of the contracting officer.

3.20.2 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where these operations are required in the original design.

3.20.3 Fastening devices. Means for preventing loss of tightness shall be provided for all screws, pins, bolts, and similar parts. Such parts, when subject to removal or adjustments, shall not be swaged, peened, staked, or otherwise permanently deformed.

3.20.4 Surfaces. All surfaces of castings, forgings, molded parts, stampings, and welded parts shall be cleaned and made free of sand, dirt, fins, sprues, scale, flux, and other harmful or extraneous materials. External surfaces shall be smooth and all edges shall be either rounded or beveled.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).

b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. First article inspection shall be applied to the preproduction sample submitted in accordance with 3.2. Failure of the first article sample to pass the examination of one or more of the first article tests shall be cause for rejection. Unless otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.3 and all the tests of 4.4.

4.2.2 Quality conformance inspection. Quality conformance inspection shall consist of examination and tests applied to each item prior to its being offered for acceptance under the contract. Unless otherwise specified (see 6.2), quality conformance inspection shall consist of all examinations of 4.3 and the test of 4.4.1.

4.3 Examination. The machine and equipment shall be examined for design, construction, materials, components, electrical equipment, and workmanship to determine compliance with the requirements of this specification. Any deviation from these requirements shall be cause for rejection.

4.4 Tests.

4.4.1 Operational test. The buffing machine shall be operated for no less than 15 minutes for observation of excessive vibration, proper operation of motor and controls, lack of lubrication, overheated bearings and capability of brush-buffing capacity in accordance with 3.7. Failure of the machine or any component to perform satisfactorily shall be cause for rejection.

4.4.2 Motor tests. The contractor shall certify that all motors have been tested and comply with the applicable test requirements of CC-M-641.

4.4.3 Test for electromagnetic interference characteristics. When conformance to electromagnetic characteristic limits is required, the equipment shall be tested in accordance with MIL-STD-462 to determine conformance to electromagnetic interference characteristic of 3.16.

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4.5 Inspection of preparation for delivery. The preservation, packaging, packing, and marking of the equipment shall be inspected to determine conformance with the applicable requirements of section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. The buffer shall be preserved and packaged in accordance with the applicable level A requirements of MIL-M-18058.

5.1.2 Level C. The buffer shall be preserved and packaged in accordance with the supplier's standard practice.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. The buffer shall be packed in accordance with the applicable level A requirements of MIL-M-18058.

5.2.2 Level B. The buffer shall be packed in accordance with the applicable level B requirements of MIL-M-18058.

5.2.3 Level C. The buffer shall be packed in a manner which will insure arrival at destination in satisfactory condition and be acceptable to the carrier at lowest rates. Containers and packing shall comply with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.3 Marking. In addition to any marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The buffing machine covered by this specification is intended primarily for use in cleaning and reconditioning plants, covered by Y & D Drawing No. 889787, to buff the exterior surfaces of 5-gallon capacity gasoline cans prior to painting.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. When first article sample is required (see 3.2).

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c. When spindle speed shall be other than as specified (see 3.7).

d. Voltage characteristics, for initial wiring of machine (see 3.11.2).

e. When electromagnetic interference control is required (see 3.16).

f. When fungus proofing is required (see 3.17).

g. Requirements for technical publications (see 3.18).

h. Spare parts required (see 3.19).

i. First article inspection, if different (see 4.2.1).

j. Quality conformance inspection, if different (see 4.2.2).

k. Level of preservation and packaging, and level of packing required (see 5.1 and 5.2).

6.3 First article. When a supplier is submitting a bid on an item identical to one that has successfully passed a first article test, he should state which agency made the procurement, the contract or order number, and the place and the date of the test. The supplier should also state the proposed charges included in his bid for such testing. If the award is made to such a supplier, the contracting officer may waive the test and reduce the bid price accordingly.

Preparing activity:
Navy - YD

(Project No. 4940-N0017Y)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

MIL-B-28524(YD) - Buffing Machine (Floor Mounted, Electric)

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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